YNU
YOKOHAMA National University

59-fJ/bit Si Photonic Crystal Slow-Light Modulator with FinFET-Compatible Driving Voltage

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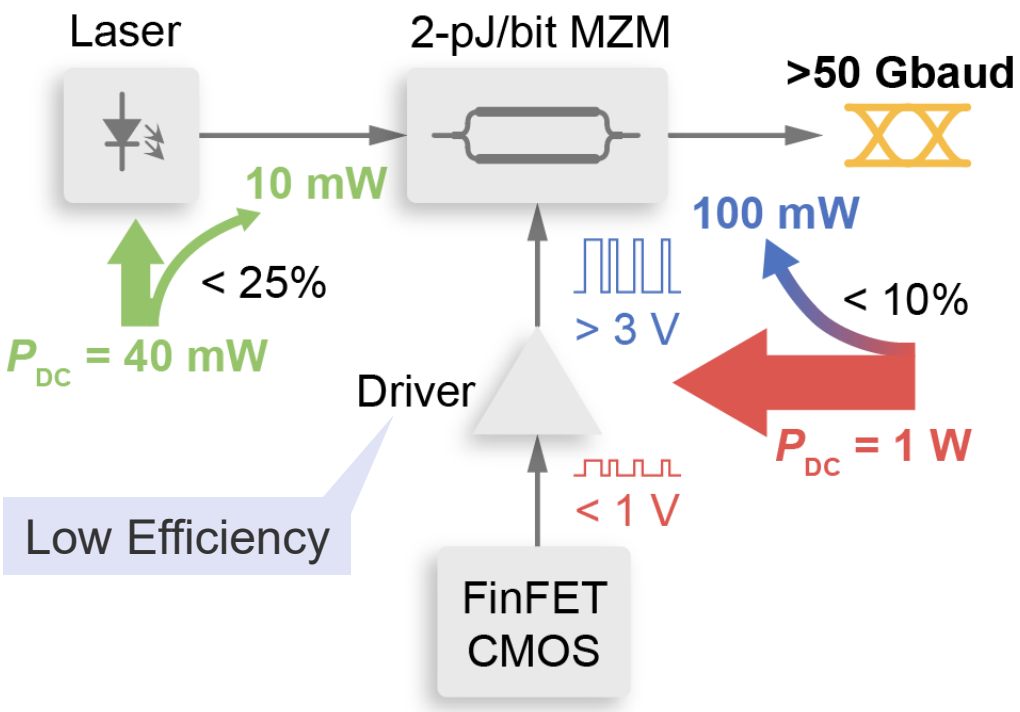
This work was supported by JSPS KAKENHI Grant Number JP23KJ0988.
This work was supported through the activities of VDEC, The University of Tokyo, in collaboration with Cadence Design Systems, Inc. and Keysight Technologies, Inc.

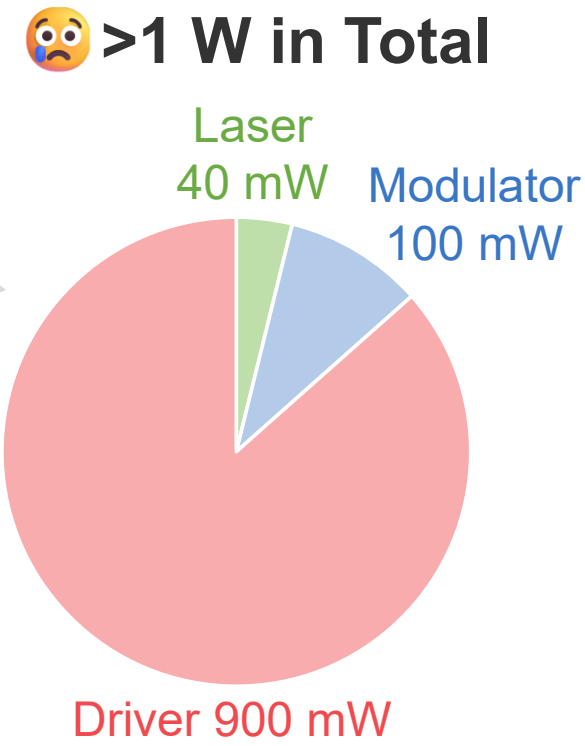
Background: Optical Interconnects

Significant increase in energy use in data centers and HPC

Necessity for high-efficiency SiPh transceivers

In SiPh transmitters, modulators and drivers consume most power



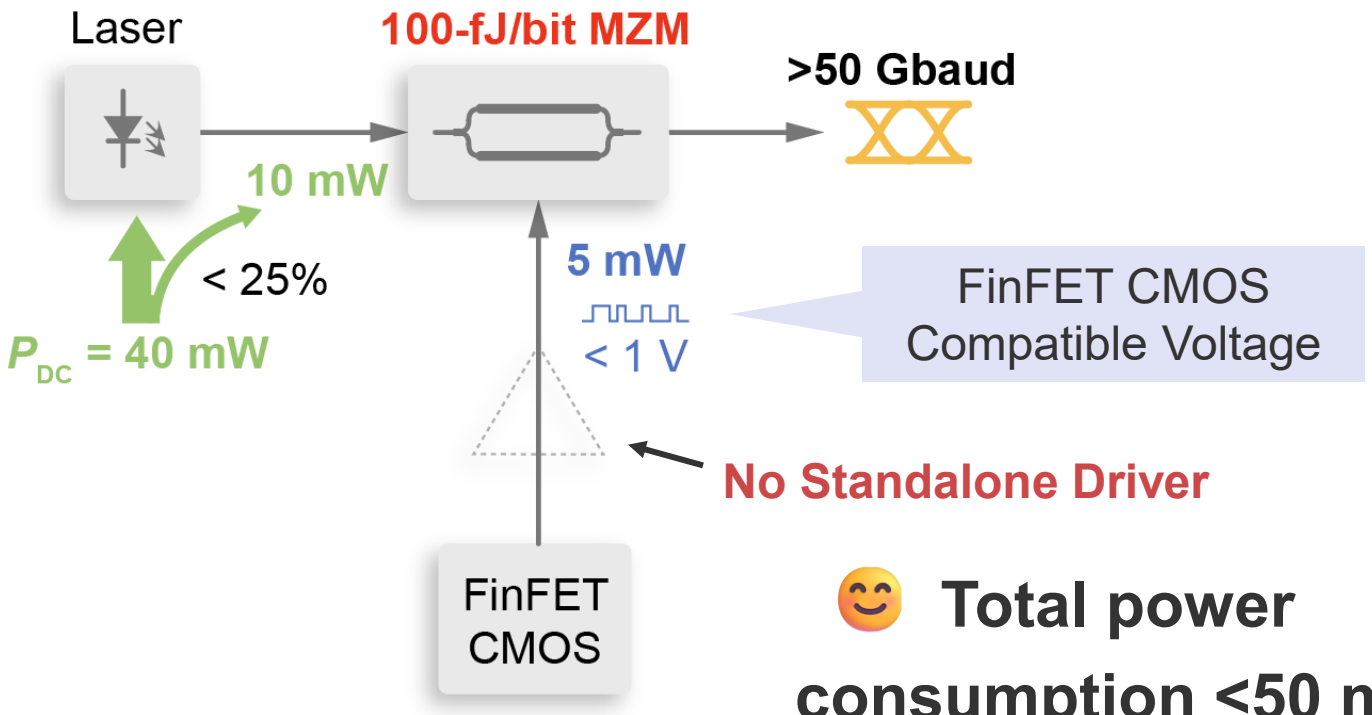


Background: Optical Interconnects

Significant increase in energy use in data centers and HPC

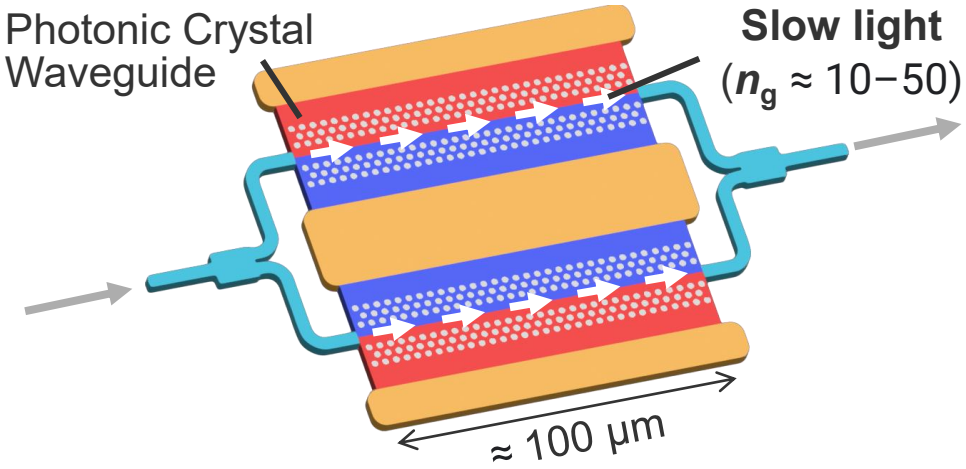
Necessity for high-efficiency SiPh transceivers

If modulator's efficiency is improved to **100 fJ/bit** ...



😊 **Total power consumption <50 mW**

Si Slow-Light MZM



Modulation efficiency ∝ n_g

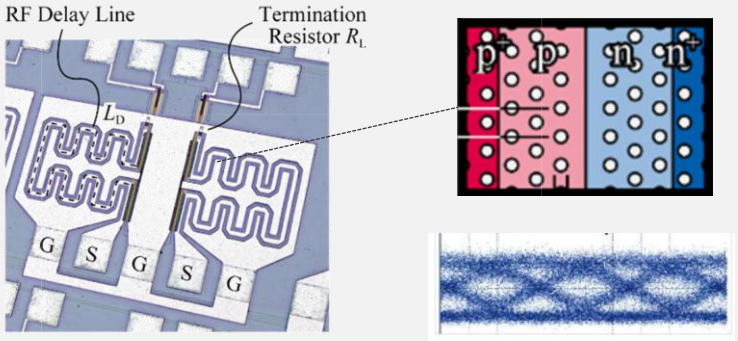
😊 Small footprint (≈ 100 μm)

😊 High speed (>64 Gbaud)

😢 High driving voltage (>3 V)

Yokohama National Univ., 2021

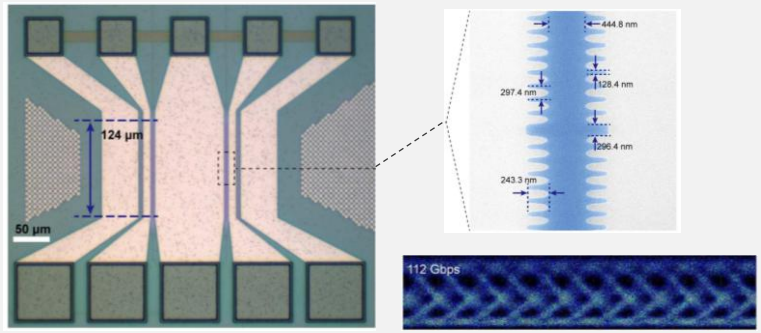
64 Gbaud V_{pp} = 3.5 V



Y. Hinakura et al., IEEE JSQTE., **27** (2021).

Peking Univ., 2023

112 Gbaud V_{pp} = 5.0 V

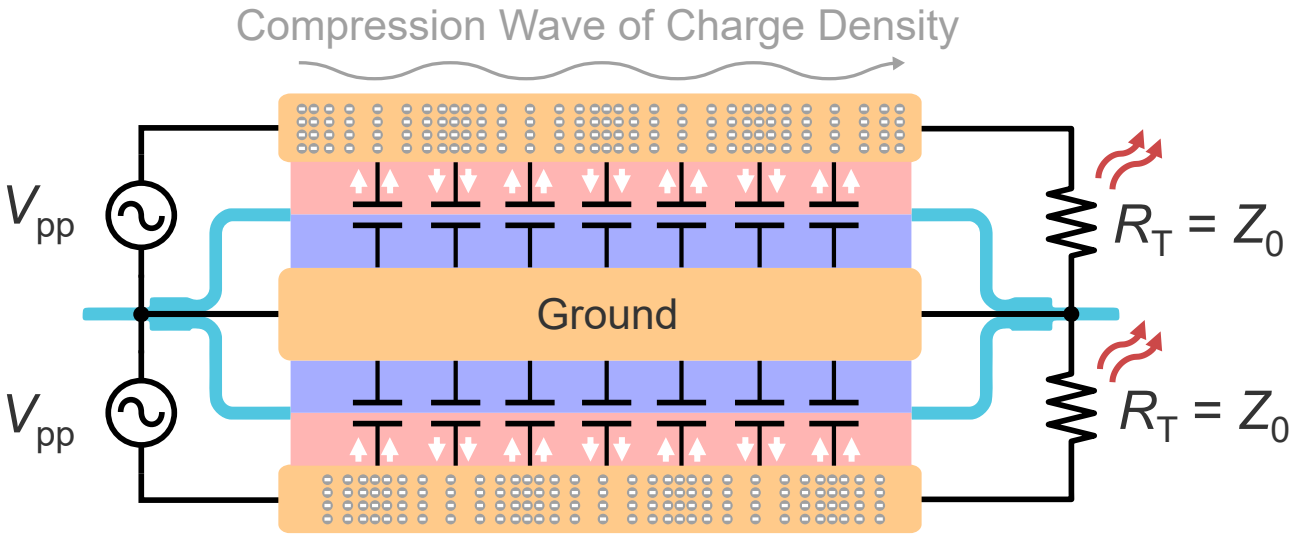


C. Han et al., Science Advances **9** (2023).

Bit energy is still limited to several pJ/bit

Challenges to High Efficiency

Most power is consumed by the termination resistor in traveling-wave electrodes



Power Consumption

$$P_{RF} = \frac{V_{pp}^2}{4R_T}$$

To reduce the power consumption, we need to

① Increase R_T, but Z_0 is typically **30 Ω** due to large C_{pn}

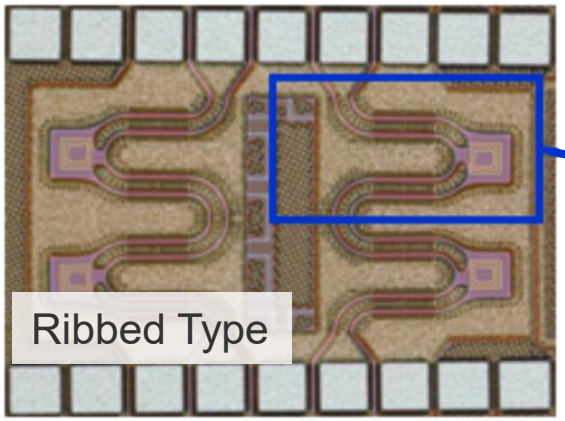
② Decrease V_{pp}, but long phase shifter or large n_g causes **EO phase mismatch**

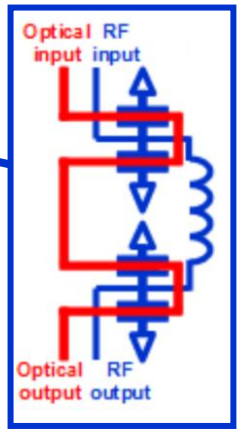
High impedance and EO phase matching are critical challenges

RF Design in Si MZMs

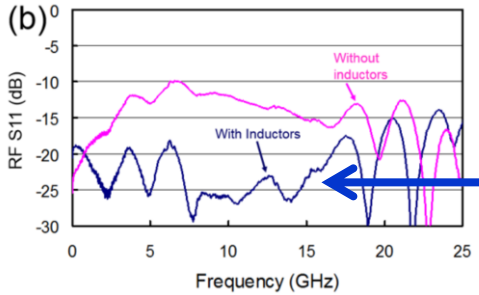
High Impedance

Distributed Electrode（IBM, 2015）





LC Ladder with Inductors



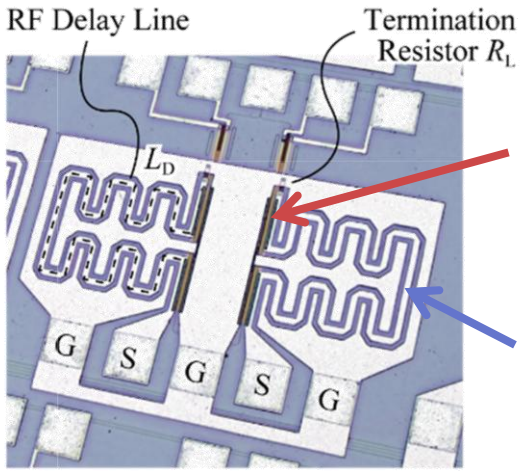
50 Ω

D. M. Gill et al., Opt. Express, **23** (2015).

😢 Large RF Loss
😢 Degraded Phase Matching

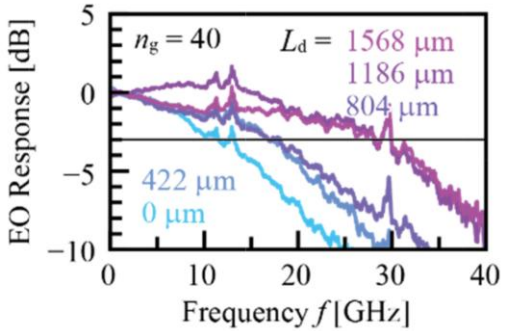
EO Phase Matching

Meander Line Electrode（YNU, 2018–）



Slow-light Phase Shifter

RF Delay Line

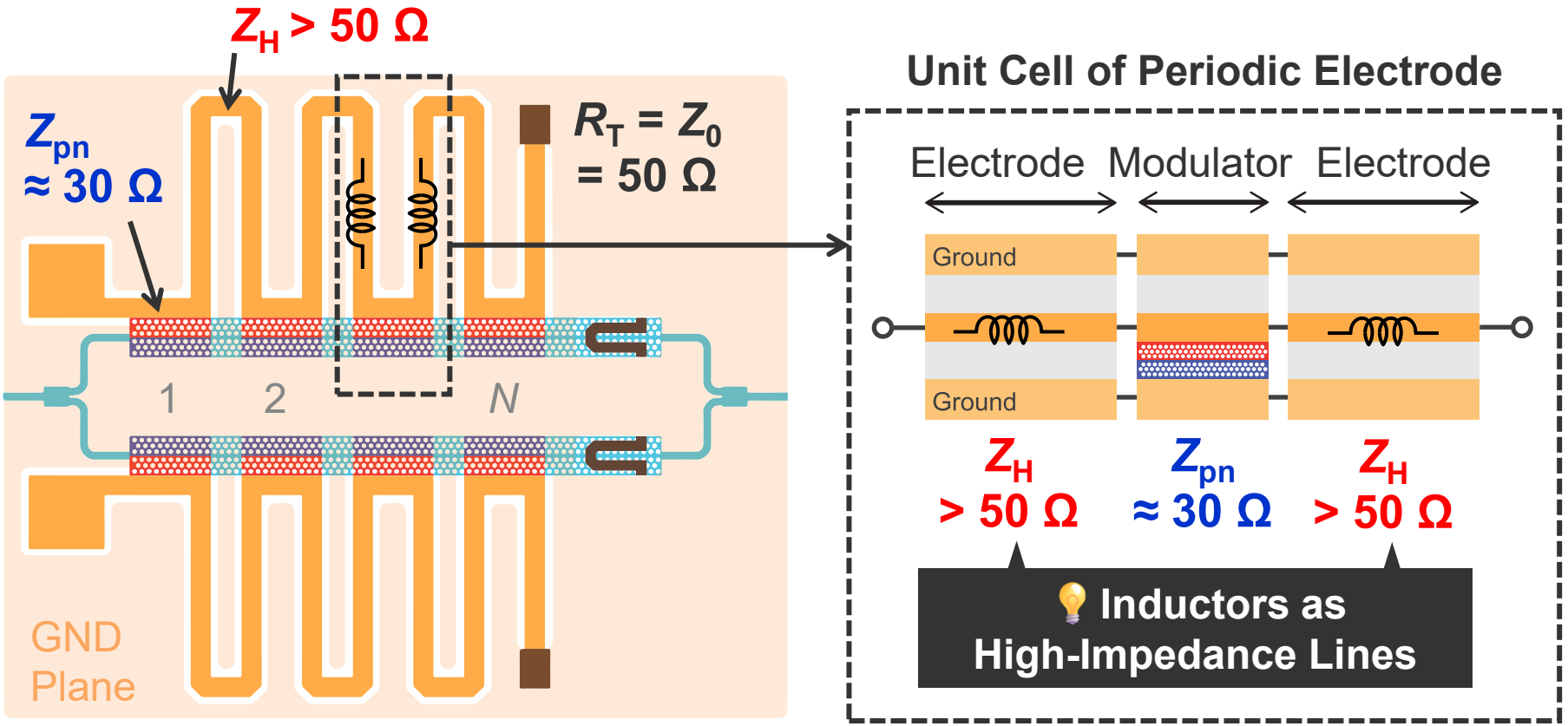


Y. Hinakura et al., IEEE JSQTE., **27** (2021)..

😢 Low Impedance

Distributed Electrode Slow-Light MZM

Periodic electrodes with segmented phase shifters and high-Z lines ($Z_H > Z_0$)

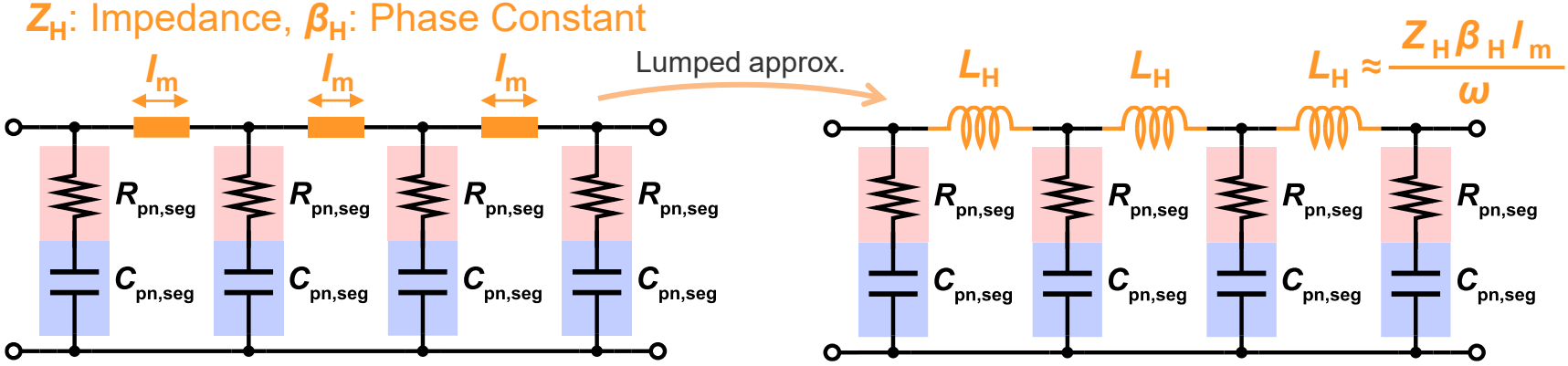


① **High impedance** and ② **EO phase matching** under low RF loss

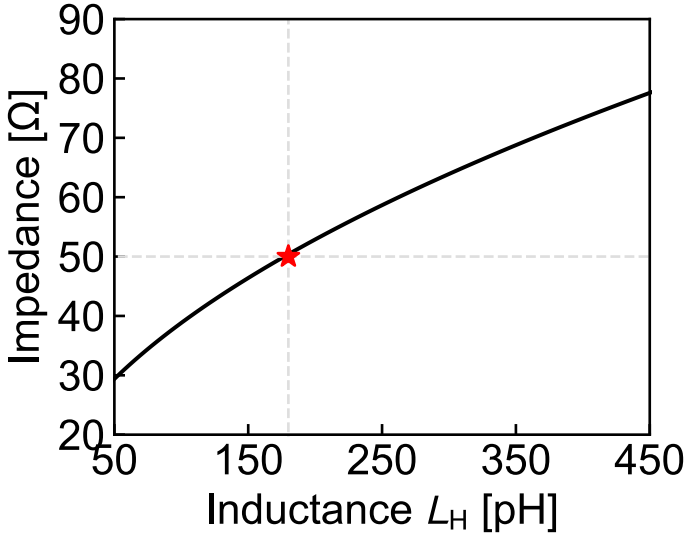
① High Impedance

Line inductance and C_{pn} form a LC ladder like a distributed constant circuit

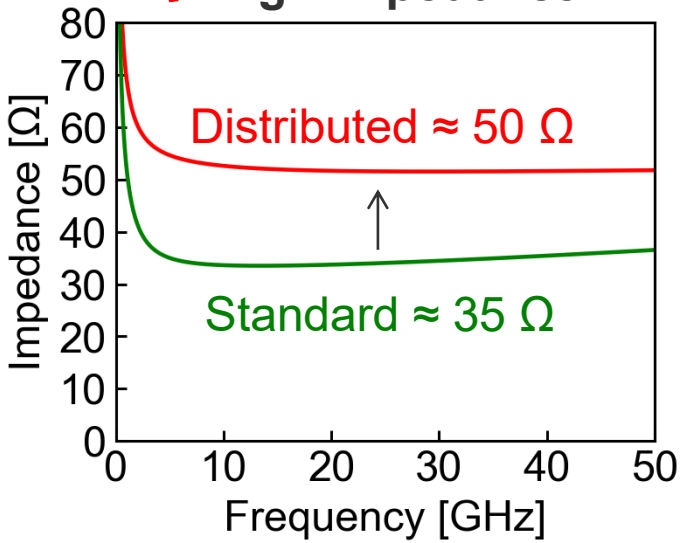
Z_H: Impedance, β_H: Phase Constant



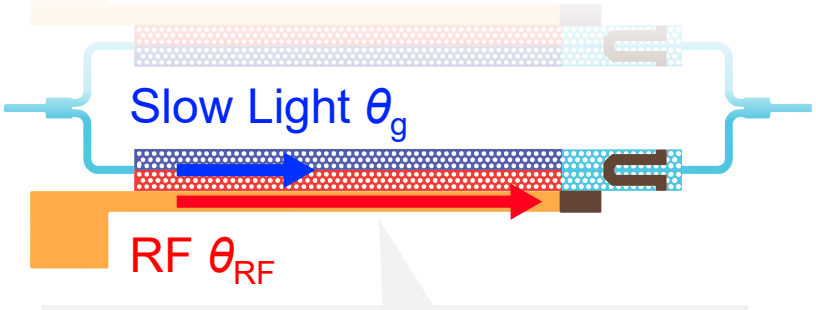
L_H increases Z_0



✓ High Impedance

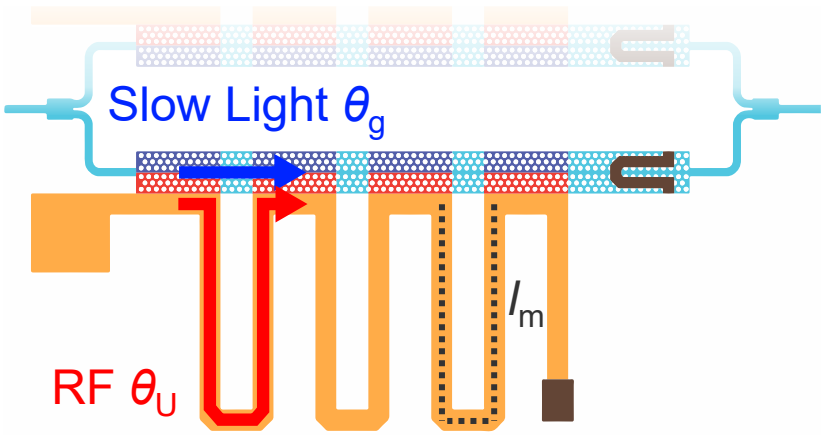


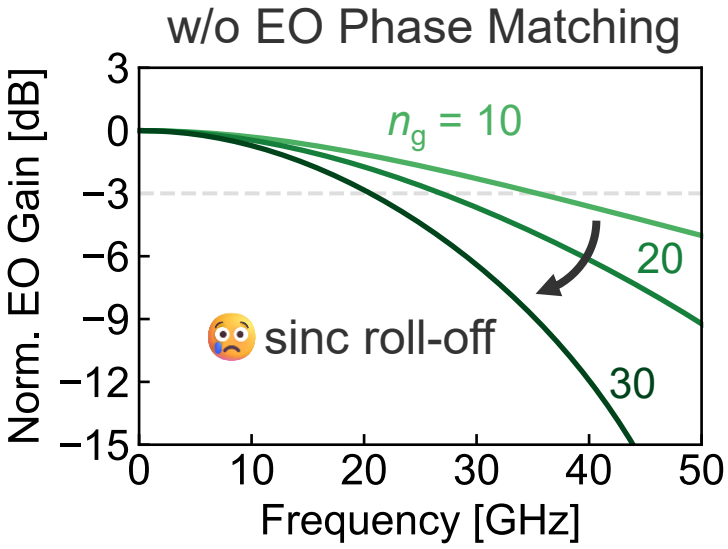
② EO Phase Matching

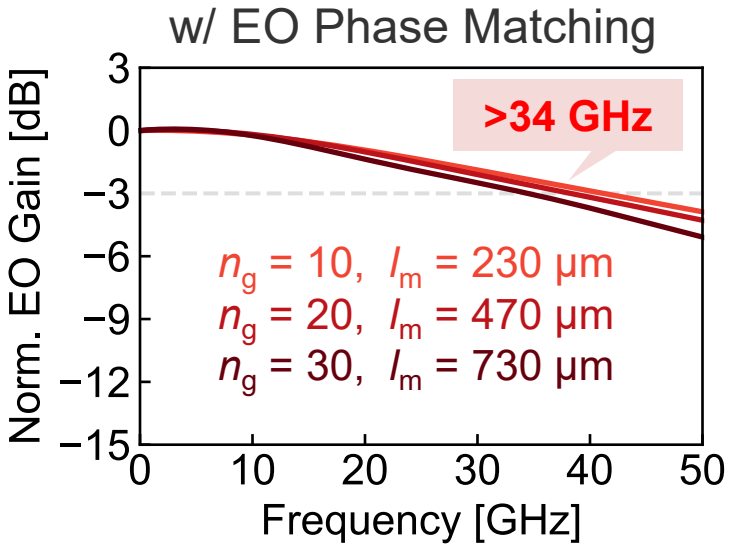


Roll-off due to phase mismatch

$\eta(f) = \mathrm{sinc}[(\theta_g - \theta_{RF}) / 2]$



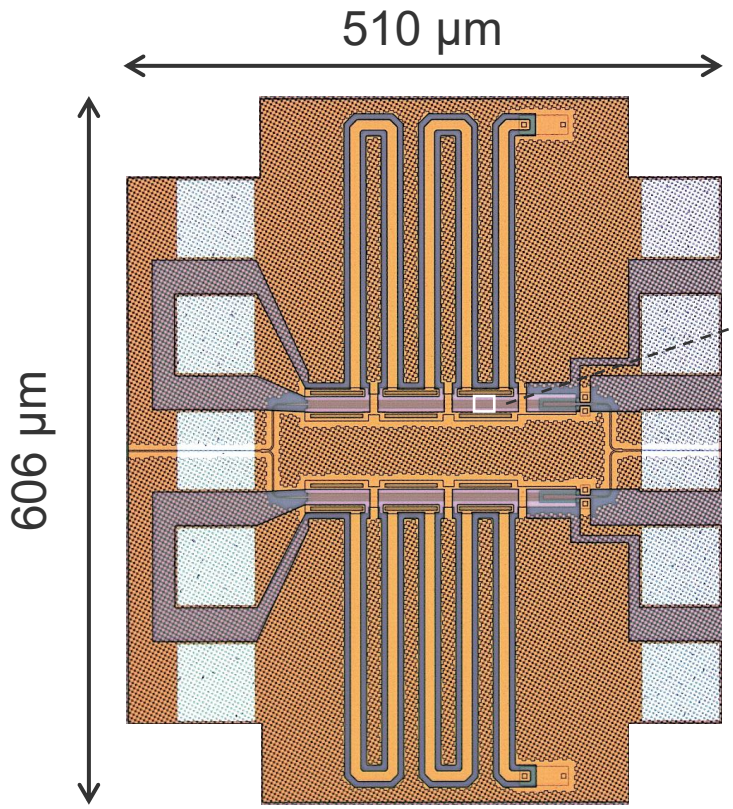


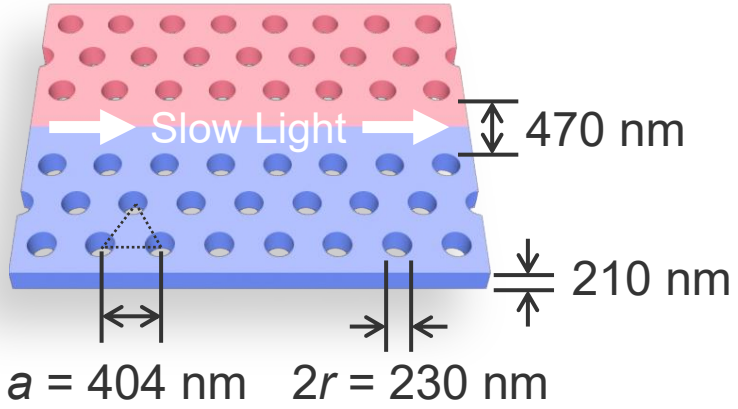


✓ Increasing n_g and reducing driving voltage while maintaining EO bandwidth

Prototype Modulator

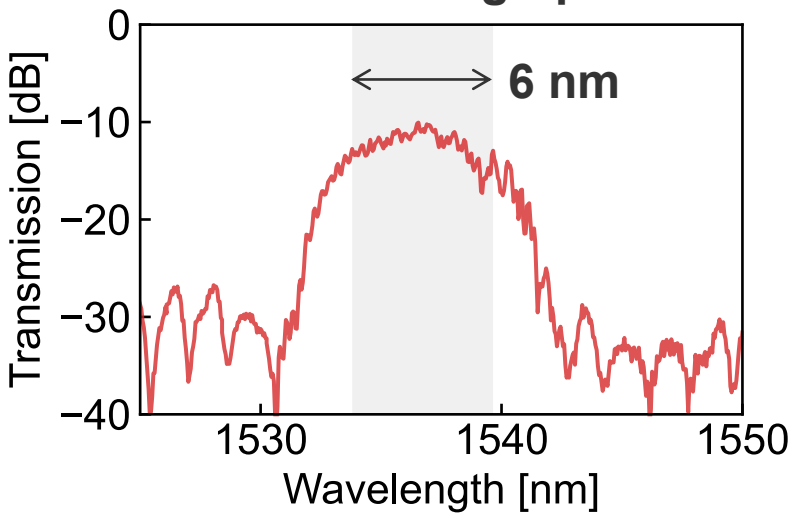
Fabricated using a CMOS-compatible 300-mm SOI wafer process



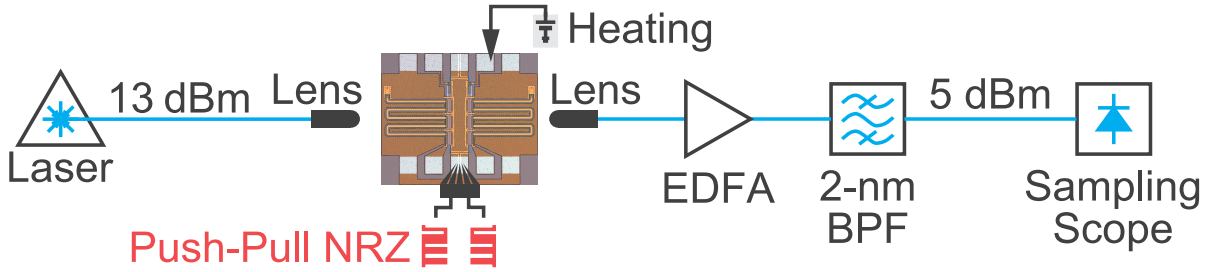


✓ **0.3-mm² Small Footprint**

✓ **Wide Working Spectrum**



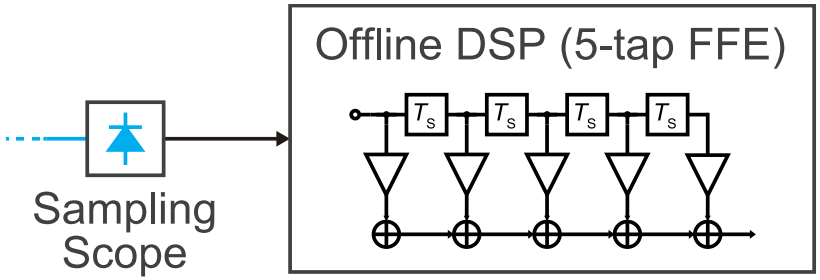
DSP-Free Signal Transmission

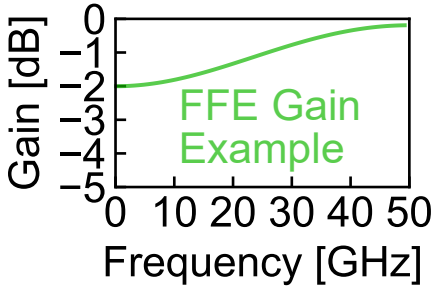


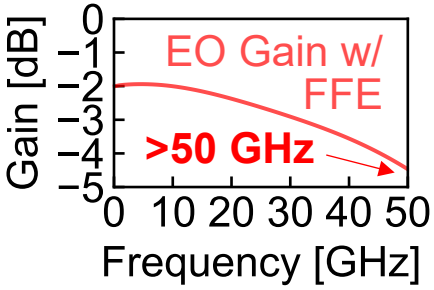
	V_{pp} = 2.2 V	1.6 V: FinFET IO	0.87 V: FinFET Core
50 GBd	493 fJ/bit *ER*: 4.5 dB, *ML*: 6.3 dB *BER* = 1.6×10^{-5} (✓HD-FEC)	246 fJ/bit *ER*: 3.0 dB, *ML*: 6.8 dB *BER* = 2.0×10^{-4} (✓HD-FEC)	76 fJ/bit *ER*: 2.0 dB, *ML*: 7.1 dB *BER* = 2.3×10^{-3} (✓HD-FEC)
64 GBd	385 fJ/bit *ER*: 4.1 dB, *ML*: 6.4 dB *BER* = 5.9×10^{-3} (✓SD-FEC)	193 fJ/bit *ER*: 2.8 dB, *ML*: 6.8 dB *BER* = 1.1×10^{-2} (✓SD-FEC)	59 fJ/bit *ER*: 1.8 dB, *ML*: 7.2 dB *BER* = 2.4×10^{-2}

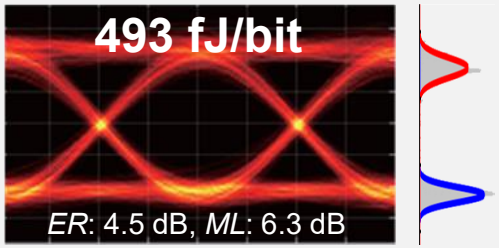
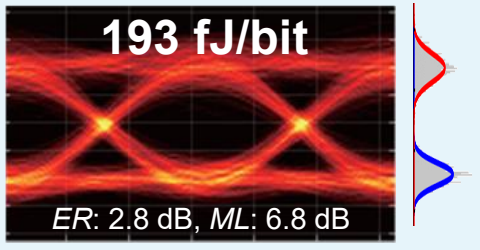
DSP-free transmission demonstrated up to 76 fJ/bit

Enhancing Symbol Rate w/ FFE







	V_{pp} = 2.2 V	**1.6 V:** FinFET IO	**0.87 V:** FinFET Core
50 GBd	 BER = 2.8×10^{-13} (✓Error Free)	 BER = 4.1×10^{-8} (✓HD-FEC)	 BER = 4.3×10^{-4} (✓HD-FEC)
64 GBd	 BER = 4.0×10^{-8} (✓HD-FEC)	 BER = 1.5×10^{-5} (✓HD-FEC)	 BER = 2.4×10^{-3} (✓HD-FEC)

Distributed electrodes w/ FFE offers the potential for >100 GBd

Conclusion

59-fJ/bit Si Photonic Crystal Slow-Light Modulator with FinFET-Compatible Driving Voltage

Purpose	Improve energy efficiency of optical modulators
Proposal	Distribute electrode slow-light Si MZM
Achievement	✓ **0.3-mm^2** small footprint
	✓ **50-Ω** flat impedance
	✓ **64-Gbaud** transmission with V_{pp} **= 0.87 V**
	✓ Energy efficiency of **59 fJ/bit**
	✓ **Potential for >100 GBd** with FFE